

Chemical Engineering for Humanity

CHEN 1000, Fall 2025

Chemical Engineering for Sustainability.

You work at a sustainability focused venture capital firm and have been tasked with doing technical diligence for a new area that your firm is interested in investing in. Your job is to provide both market and technical information to the partners at the firm so they can properly evaluate pitches from start-up companies. In this project, you are tasked with evaluating the market for bio-polyesters (e.g. poly(lactic acid), poly(hydroxyalkanoates)) and how switching to these materials could provide a more sustainable alternative to petroleum-derived plastics on the market today. Answer the following questions, carefully document the answers, and provide your guidance to the investment team.

1. What current plastics could these bio-based (or biosynthesized) polyesters replace? How many tons of the commodity plastics to be replaced and bio-plastic alternatives are produced annually? What is the market size for the polymers that bio-polyesters could displace? What types of items can be made from bio-polyesters?
2. Estimate how much PLA can be made from corn. Assume you want to displace 100% of the current polyester market, how many tons of corn will be needed to produce enough PLA to do so? What percent of current corn production will this require? How much land area is required to produce this much corn?
3. Identify potential advantages and disadvantages (related to sustainability) of switching to bio-polyesters. Highlight potential areas for improvement of bio-polymer alternatives that could be attractive for a start-up to tackle.